

Carlos D. Martinez

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Professional Summary

Detail-oriented Computer Science student with a passion for Hardware engineering and full-stack development. Experienced in building web applications, cloud-integrated machine learning pipelines, and scripting automation tools. Proficient in Python, Java, C++, and JavaScript, with hands-on skills in Git, Docker, Azure, SQL, and PCB. Highly skilled at writing clean, maintainable code and collaborating with cross-functional teams. Actively seeking a software engineering role to contribute to impactful technology projects.

Education

San Diego State University, San Diego, CA

B.S.in Computer Science

May 2025

Relevant Coursework: Artificial Intelligence, Database Theory & Implementation, Robotics, Operating Systems, Software Engineering

Universidad Autónoma de Baja California (UABC), Mexicali, B.C., Mexico

Engineering in Electronics (5 semesters completed)

2016–2018

Relevant Coursework: Electricity & Magnetism, Differential Equations, Circuits I & II, Multivariable Calculus, Modern Physics, Advanced Programming in LabVIEW, Dynamics, Thermodynamics, Statics

CBTIS 21, Mexicali, B.C., Mexico

Associate's in Electronics

May 2016

Relevant Coursework: Microcontroller Applications for Smart Buildings, PLC Programming, Digital Design, Analog Circuits and Radiofrequency

Work Experience

Case Manager

Self Determine Futures LLC, San Diego, CA

Mar 2025 - Present

- Write session notes, plans of action, progress summaries
- Translate complex situations into clear, organized documentation

STEM Instructor

Nexplore, San Diego, CA

Sep 2024 - Aug 2025

- Teach hands-on classes in drone building, 3D printing, and engineering to K-8 students.

- Coordinate with schools and curriculum designers to deliver engaging STEM content.

Unix/Linux System Administration Internship

San Diego State University, San Diego, CA

Feb 2025 - May 2025

- Built a distributed Linux-based network with DNS, Mail, Web, and Authentication services.
 - Used Kubernetes to deploy and load-balance redundant web servers.
 - Managed automation with Ansible scripts and implemented LDAP user management
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Projects

Azure ML: Diabetes Prediction Model Comparison

Azure Machine Learning, Classification Models, Web UI

- Created an Azure Machine Learning workspace and uploaded a public diabetes dataset.
- Launched and configured training jobs to compare Logistic Regression, Random Forest, and Neural Network models.
- Evaluated metrics like accuracy and AUC to select the best-performing model.
- Analyzed results visually using Azure Studio and tracked outputs via web interface.

House Price Prediction Model

Python, Pandas, Scikit-learn, Regression Models

- Preprocessed large housing datasets across multiple states using pandas.
- Applied Linear Regression, Random Forest, and Deep Forest models.
- Tuned features to improve model accuracy from 72% to 86%; visualized results with matplotlib.

Website: <https://davidmtzh.github.io/Portfolio-Website/>

HTML, CSS, JavaScript, GitHub Pages

- Designed and deployed a responsive multi-page website for my portfolio.
 - Implemented mobile-first styling and visual branding with clean UI/UX practices.
 - Hosted via GitHub Pages with custom domain setup and NGINX proxy.
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Technical Skills

- **Languages:** Python, SQL, C++, Java, C#, Bash, Haskell, Assembly
- **Cloud & DevOps:** Microsoft Azure, Kubernetes, Azure Machine Learning, Git, Docker, Linux, Ansible

- **Libraries/Frameworks:** scikit-learn, pandas, matplotlib, TensorFlow NumPy
- **Data Science:** Data cleaning, feature engineering, model evaluation, data analysis, Jupiter notebooks
- **Web Dev:** HTML, CSS, JavaScript, GitHub Pages, NGINX

